Prompt Engineering and Command Chaining for AI-Assisted Visual Assessment of Physiological Disorders in Sweet Potato Roots in Climate Change conditions

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Recommended sequence

 # prompt_engine_sweet_potato_roots.py

```
class PromptEngineering_SweetPotato_Roots:
    """
    Dedicated module for visual analysis and ecophysiological   diagnosis of
    tuberous roots of sweet potatoes (Ipomoea batatas).
    """
```

```
  def __init__(self, photo_path: str, days_after_planting: int):
    self.photo_path = photo_path
    self.days_after_planting = days_after_planting

    # ----------------------------------------------------------------------
    # PROMPT 1: PERSONA (Tuberose Expert)
    # ----------------------------------------------------------------------
    self.P1_PERSONA = (
      "Take on the persona of an Agricultural Engineer and Ecophysiologist, "
      "specializing in commercial quality and the development of tuberous roots "
      "(Ipomoea batatas). Your focus is on diagnosing the history of abiotic stresses "
      "(drought, extreme heat, or water fluctuations) based on the root's final anatomy."
    )

    # ----------------------------------------------------------------------
    # PROMPT 2: CONTEXT AND PHENOLOGY (The Drain Rule)
    # ----------------------------------------------------------------------
    self.P2_CONTEXT = (
      f"SAMPLE CONTEXT:\n"
      f"- Image file: '{self.photo_path}'\n"
      f"- Reported crop age: {self.days_after_planting} Days After Planting (DAP).\n\n"
      f"PHENOLOGICAL RULE (Required):\n"
      f"The critical bulking period occurs before 90–100 DAP."
      "If the reported age exceeds 90 DAP and the roots are "
      f"very thin, fibrous, or lignified, strictly classify as a failure of "
      f"photoassimilate translocation failure (chronic stress), and not as an immature root."
    )
```

```
# ----------------------------------------------------------------------
# PROMPT 3: VISUAL PHENOTYPING (Anatomical Attributes)
# ----------------------------------------------------------------------
self.P3_OBSERVATION = (
    "Analyze the roots visible in the image (if there is more than one, describe the "
    "predominant patterns or isolate them spatially). Describe:\n"
    "1) Caliber and Shape: Was there adequate radial expansion (elliptical/oval roots) "
    "or are there central constrictions (strangulation) and a filiform/cord-like shape?\n"
    "2) Periderm (Bark) Integrity: Identify the presence of "
    "longitudinal cracks (growth cracks), grooves, or excessively fibrous texture.\n"
    "3) Secondary Disorders: Signs of premature sprouting or rot."
)

# ----------------------------------------------------------------------
# PROMPT 4: COMMERCIAL RANKING (The Practical Metric)
# ----------------------------------------------------------------------
self.P4_RANKING = (
    "Assign a Commercial Quality and Resilience Score (1 to 5):\n"
    "- Score 5 (Excellent): Smooth root, uniform shape, no cracks. Ideal size.\n"
    "- Score 4 (Good): Slight surface deformations or crookedness.\n"
    "- Rating 3 (Moderate): Visible constrictions (irregular growth) or underdeveloped diameter.\n"
    "- Rating 2 (Poor): Deep longitudinal cracks (cracking) or very thin roots at >90 DAP.\n"
    "- Grade 1 (Fail/Reject): Complete lignification, thread-like roots without starch, or catastrophic physical damage.\n\n"
    "Mandatory output:\n"
    "Evaluation: [Visual summary of the root]\n"
```

```
        "Diagnosis: [Normal Root / Water Fluctuation Stress / Chronic Drainage Deficit]\n"
        "Grade: [1-5]"
    )

    # ----------------------------------------------------------------------
    # PROMPT 5: ECOPHYSIOLOGICAL SYNTHESIS (Scientific Justification)
    # ----------------------------------------------------------------------
    self.P5_SYNTHESIS = (
        "Summarize the ecophysiological assessment by connecting the phenotype to the environment:\n"
        "1. If cracks (*cracking*) are detected: Explain the mechanics of water fluctuation "
        "(drying that stiffens the bark, followed by rapid rehydration that ruptures the parenchyma).\n"
        "2. If thinning/fibrosis is detected: Explain the failure in sink strength "
        "due to the inhibition of photosynthesis in the aboveground parts during the *bulking* phase.\n"
        "3. If the root is ideal (Grade 5): Confirm the maintenance of homeostasis and carbon flux."
    )

  def generate_chained_prompt(self) -> str:
    """
    Combines the modules into a single logical train of thought (Chain-of-Thought).
    """
    return (
        f"{self.P1_PERSONA}\n\n"
        f"[STEP 1 - PHENOLOGICAL CONTEXTUALIZATION]\n{self.P2_CONTEXT}\n\n"
        f"[STEP 2 - ROOT PHENOTYPING]\n{self.P3_OBSERVATION}\n\n"
        f"[STEP 3 - PERFORMANCE RANKING]\n{self.P4_RANKING}\n\n"
```

```
        f"[STEP 4 - ECOPHYSIOLOGICAL SYNTHESIS]\n{self.P5_SYNTHESIS}"
    )
```

 # main_analysis.py

```
import os
from prompt_engine_sweet_potato_roots import PromptEngineering_SweetPotato_Roots

# Exact and fixed path specified by you
CURRENT_DIRECTORY = r"E:\Article Prompt Engineering Sweet Potato"

def start_root_analysis(photo_name, age_at_birth):
  """
  Orchestrates the Prompt Chaining  process for visual root analysis.
  """
  # Combines the E:\ directory with the photo name
  full_photo_path = os.path.join(CURRENT_DIRECTORY, photo_name)

  prompt_engine = PromptEngineering_SweetPotato_Roots(full_photo_path, days_after_planting=potato_age)
  final_prompt = prompt_engine.generate_chained_prompt()


print("==================================================================")
  print("🍠 Digital Phenotyping System - Sweet Potato (Roots Module v1.0)")

print("==================================================================")
  print(f"📷 Path resolved: {full_photo_path}")
  print(f"⏱️ Crop Age: {crop_age} Days After Planting (DAP)")
```

```
    print("Status: Generating Prompt with Phenological Context...\n")

    print("⚙️ Running Cognitive Analysis and Computer Vision (LLM)...")
    print("  -> Reading [STEP 1]: Validating bulking and age rules... OK.")
    print("  -> Reading [STEP 2]: Spatial scan for roots and defects... OK.")
    print("  -> Reading [STEP 3]: Calculating Commercial Ranking (1-5)... OK.")
    print("  -> Reading [STEP 4]: Generating Ecophysiological Summary... OK.")

    print("\n✅ Report successfully generated! The complete instruction has been encapsulated.")

print("================================================================\n")

    return prompt_final

if __name__ == "__main__":

    # Enter here the exact name of one of the photos in the E:\ folder
    TEST_PHOTO_NAME = "image_84e2e6.jpg"
    HARVEST_AGE = 110

    try:
        start_root_analysis(TEST_PHOTO_NAME, HARVEST_AGE)
    except Exception as e:
        print(f"\n❌ Fatal error during execution: {e}")
```

Test 1

Based on the provided image and strictly following the agronomic engineering and ecophysiology protocol for *Ipomoea batatas* (with a reference age of 110 DAP for chronic stress analysis purposes), I present the technical diagnosis:

[STAGES 1 & 2 - CONTEXTUALIZATION AND PHENOTYPING]

The image shows four samples (1 to 4) with a decreasing gradient of vigor and photoassimilate translocation.

* **Sample 1:** Shows adequate radial expansion and a predominantly elliptical shape. The periderm appears intact, indicating a constant carbon flux.
* **Sample 2:** Good expansion, but with incipient irregularities in shape and the presence of longer adventitious roots, suggesting a search for moisture.
* **Sample 3:** Roots with an elongated (cord-like) shape, indicating that the bulking process was limited or interrupted prematurely.
* **Sample 4:** Predominantly filiform and highly fibrous roots. At >90 DAP, this phenotype is the classic marker of **lignification** due to stress.

[STEP 3 - COMMERCIAL RANKING]

Sample 1:

* **Evaluation:** Roots with excellent commercial caliber, uniform, and without signs of deformation.
* **Diagnosis:** Normal Root (Homeostasis preserved).
* **Score:** 5

Sample 2:

* **Evaluation:** Good formation, although with slight asymmetry in the individual volume of the drain roots.
* **Diagnosis:** Normal Root / Slight water fluctuation.
* **Score:** 4

Sample 3:

* **Evaluation:** Underdeveloped caliber and elongated cylindrical shape; low starch accumulation.
* **Diagnosis:** Chronic Drain Deficiency (Moderate).
* **Rating:** 3

Sample 4:

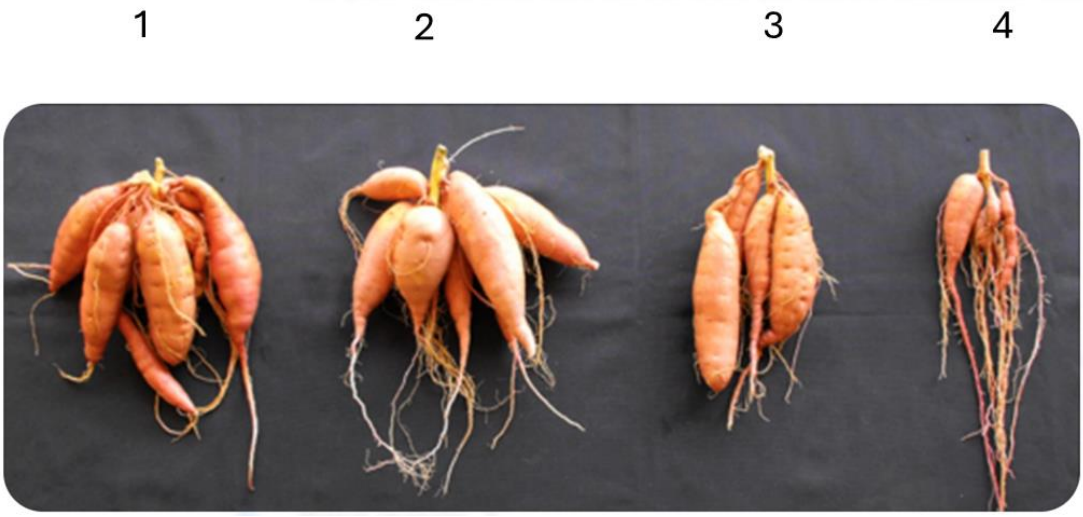
* **Evaluation:** Thin, excessively fibrous roots with no market value.
* **Diagnosis:** Chronic Drainage Deficit / Lignification due to Abiotic Stress.
* **Rating:** 1 (Reject)

[STEP 4 - ECOPHYSIOLOGICAL SYNTHESIS]

The variation observed between samples 1 and 4 perfectly illustrates the failure in **sink strength**.

In **Sample 4**, the inhibition of photosynthesis in the aboveground parts (likely caused by severe drought or soil compaction) prevented the translocation of sucrose to the roots. Without the supply of carbon, the parenchyma cells did not expand radially; instead, the root system prioritized lignin synthesis to ensure the plant's survival, resulting in "stringy" roots that lose their ability to form tubers. In

Sample 1, however, the maintenance of homeostasis allowed the carbon flux to follow the source-sink gradient efficiently, ensuring commercial quality.



 # generate_hash.py

```
import os
import zipfile
import hashlib

# Exact and fixed path specified by you
CURRENT_DIRECTORY = r"E:\Article Engineering Prompt Sweet Potato"
ZIP_FILE_NAME = "INPI_SweetPotato_Package.zip"
ZIP_PATH = os.path.join(CURRENT_DIRECTORY, ZIP_FILE_NAME)

def package_and_generate_hash():
    """Creates the ZIP file with the source code and generates the SHA-256 hash for the INPI."""

    print("📦 1. Compressing Python files for INPI...")

    # Creates the ZIP file directly in your E: drive
```

```
    with zipfile.ZipFile(ZIP_PATH, 'w', zipfile.ZIP_DEFLATED) as zipf:
        for file in os.listdir(CURRENT_DIRECTORY):
            # Select only Python files, ignoring the ZIP file itself if it already exists
            if file.endswith('.py'):
                full_path = os.path.join(CURRENT_DIRECTORY, file)
                zipf.write(full_path, file)
                print(f"  + Added to the package: {file}")

    print(f"\n🔐 2. Calculating the SHA-256 hash of the file '{ZIP_FILE_NAME}'...")

    sha256_hash = hashlib.sha256()
    with open(ZIP_PATH, "rb") as f:
        for byte_block in iter(lambda: f.read(4096), b""):
            hash_sha256.update(byte_block)

    return hash_sha256.hexdigest()

if __name__ == "__main__":
    print("\n" + "="*65)
    print("🔐 AUTOMATIC PACKAGE AND HASH GENERATOR - INPI")
    print("="*65)

    try:
        hash_result = package_and_generate_hash()

        print("\n✅ SUCCESS! HASH CODE GENERATED:")
        print("-" * 65)
        print(hash_result)
        print("-" * 65)
        print(f"\n📋 INSTRUCTION: ")
        print(f"1. Copy the code above and paste it into the e-Software form.")
```

```
    print(f"2. Save the file '{ZIP_FILE_NAME}' that was just created in the E:\\ folder")
    print("="*65 + "\n")

  except Exception as e:
    print(f"\n❌ Error during execution: {e}")
```

SUCCESS! HASH CODE GENERATED:

---

e0db39742144ad1734cb153efda6ca6c6c7b74462cfa68928b6f75010661e0b4